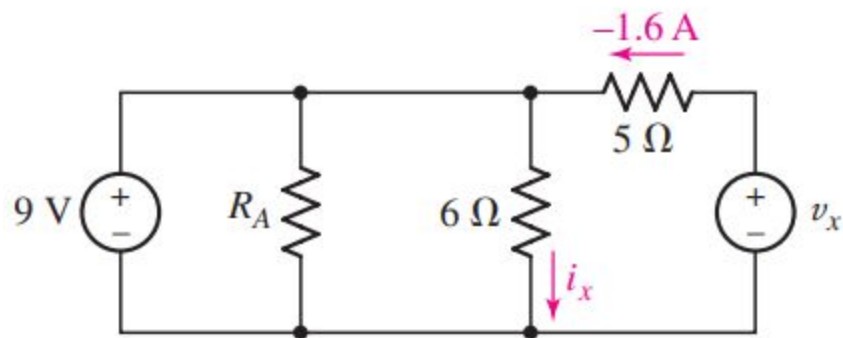


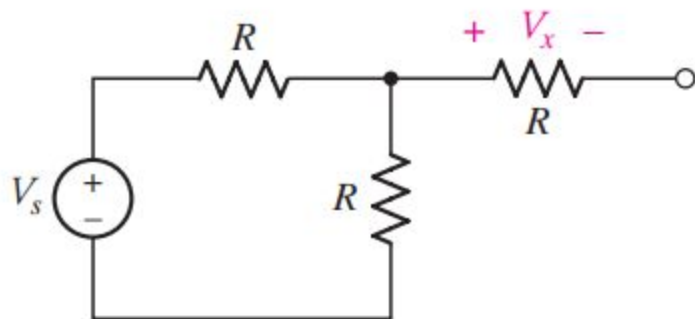
HW2: Use KCL and Ohm's law to solve the circuits

11. In the circuit depicted in Fig. 3.53, i_x is determined to be 1.5 A, and the 9 V source supplies a current of 7.6 A (that is, a current of 7.6 A leaves the positive reference terminal of the 9 V source). Determine the value of resistor R_A .

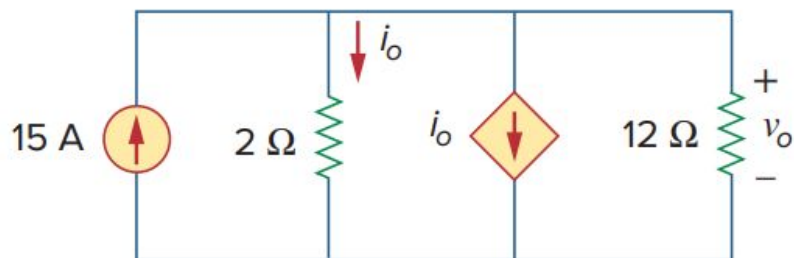


■ FIGURE 3.53

14. Study the circuit depicted in Fig. 3.56, and explain (in terms of KCL) why the voltage labeled V_x must be zero.

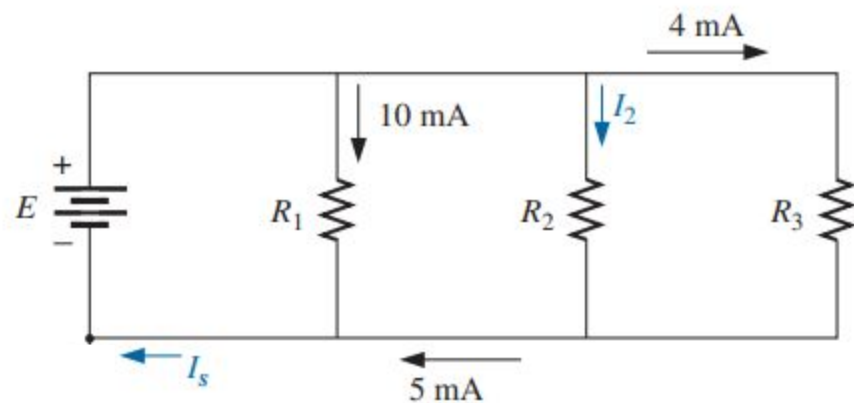


■ **FIGURE 3.56**

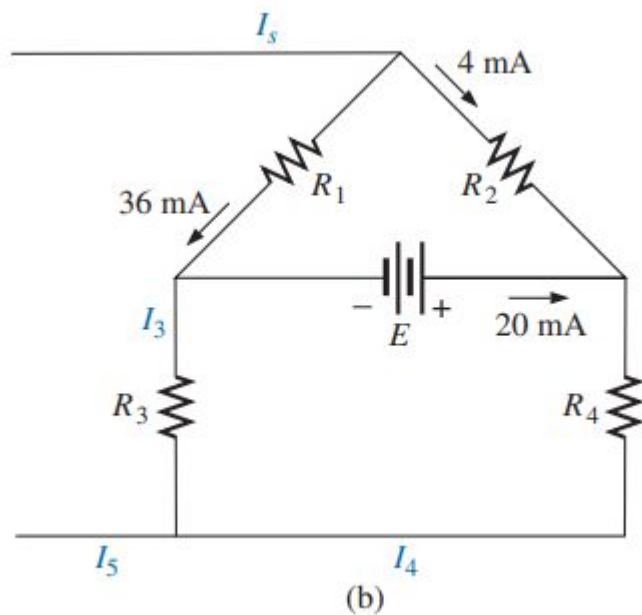


Find v_o and i_o in the circuit of Fig. 2.26.

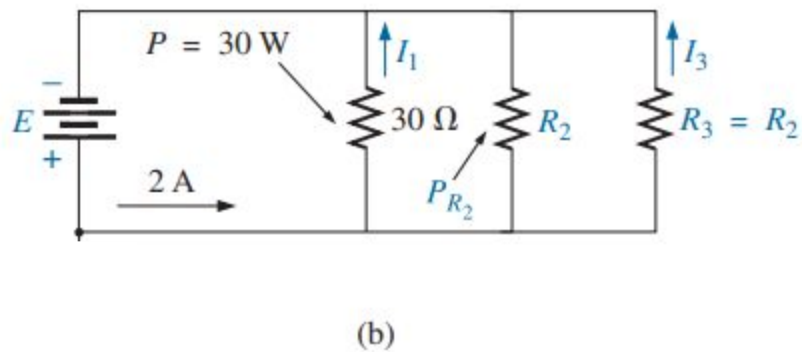
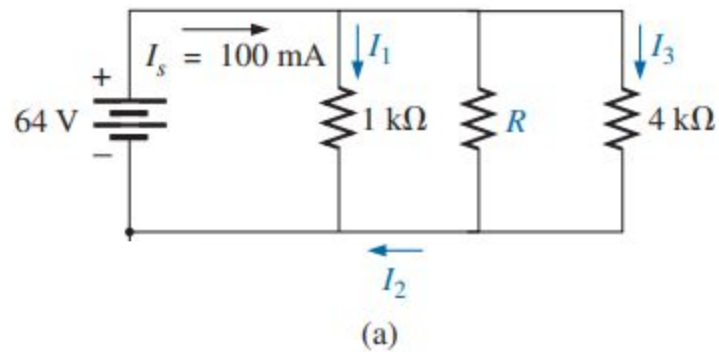
27. Using Kirchhoff's current law, determine the unknown currents for the parallel network in Fig. 6.86.



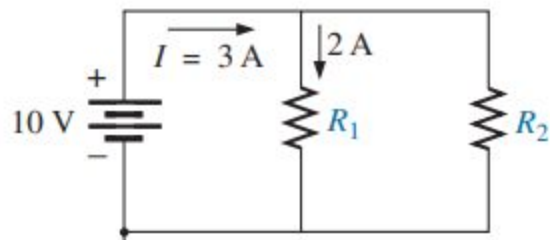
29. Using Kirchhoff's current law, determine the unknown currents for the networks in Fig. 6.88.



32. Find the unknown quantities for the networks of Fig. 6.91 using the information provided.



31. Find the unknown quantities for the networks in Fig. 6.90 using the information provided.



(a)

